One Numerical Example Solved by IRR, PW, and AW

(Mutually Exclusive Alternatives)

Given. The minimum attractive rate of return is MARR = 10% per year. Two mutually exclusive alternatives (choose *one* of them) are:

	$t=0$	1	2	3	4	5
Alternative A	−10,000	+3,000	+3,000	+3,000	+3,000	+3,000
Alternative B	−16,000	+4,200	+4,200	+4,200	+4,200	+6,200

(For B, the year-5 amount includes a salvage value of 2,000 in addition to the 4,200 annual net benefit.)

1) IRR (Rate of Return) Method

Alternative A. The IRR solves

$$0 = -10{,}000 + 3{,}000(P/A, i, 5).$$

Using a calculator or spreadsheet,

$$\boxed{\text{IRR}_A \approx 15.238\%}$$

so A is acceptable since $\text{IRR}_A > \text{MARR}$.

Alternative B. The IRR solves

$$0 = -16{,}000 + 4{,}200(P/A, i, 5) + 2{,}000(P/F, i, 5).$$

Using a calculator or spreadsheet,

$$\boxed{\text{IRR}_B \approx 12.655\%}$$

so B is also acceptable since $\text{IRR}_B > \text{MARR}$.

Important note (mutually exclusive case). Because A and B are mutually exclusive, comparing IRR_A and IRR_B is *not* sufficient for the final choice. The correct IRR-based comparison is the *incremental IRR*.

Incremental (B–A) cash flow series:

$$\Delta CF = CF_B - CF_A = (-6{,}000,\ 1{,}200,\ 1{,}200,\ 1{,}200,\ 1{,}200,\ 3{,}200).$$

Incremental IRR solves

$$0 = -6{,}000 + 1{,}200(P/A, i, 4) + 3{,}200(P/F, i, 5),$$

and (calculator/spreadsheet)

$$\boxed{\text{IRR}_{B-A} \approx 8.821\%}.$$

Since $\text{IRR}_{B-A} < \text{MARR}$, the extra investment required for B over A is not justified; therefore IRR logic selects **A**.

2) Present Worth (PW) Method at MARR = 10%

Use factor values at $i = 10\%$, $n = 5$:

$$(P/A, 10\%, 5) = 3.7908, \qquad (P/F, 10\%, 5) = 0.6209.$$

PW of A:

$$PW_A = -10{,}000 + 3{,}000(P/A, 10\%, 5) = -10{,}000 + 3{,}000(3.7908) \approx \boxed{1{,}372.36}.$$

PW of B:

$$\begin{aligned} PW_B &= -16{,}000 + 4{,}200(P/A, 10\%, 5) + 2{,}000(P/F, 10\%, 5) \\ &= -16{,}000 + 4{,}200(3.7908) + 2{,}000(0.6209) \approx \boxed{1{,}163.15}. \end{aligned}$$

Decision by PW (mutually exclusive): Choose the alternative with the larger PW:

$$PW_A > PW_B \quad \Rightarrow \quad \boxed{\text{Select A}}.$$

3) Annual Worth (AW) Method at MARR = 10%

Convert PW to an equivalent uniform annual amount using $(A/P, 10\%, 5) = 0.2638$:

$$(A/P, 10\%, 5) = \frac{i(1+i)^n}{(1+i)^n - 1} = 0.2638.$$

AW of A:

$$AW_A = PW_A(A/P, 10\%, 5) = 1{,}372.36(0.2638) \approx \boxed{362.03 \text{ per year}}.$$

AW of B:

$$AW_B = PW_B(A/P, 10\%, 5) = 1{,}163.15(0.2638) \approx \boxed{306.84 \text{ per year}}.$$

Decision by AW (mutually exclusive): Choose the alternative with the larger AW:

$$AW_A > AW_B \quad \Rightarrow \quad \boxed{\text{Select A}}.$$

Final Conclusion (All Three Methods)

- **Screening:** Both alternatives have IRR above MARR, so both are individually acceptable.
- **Correct choice (mutually exclusive):** PW and AW select **A**. Incremental IRR also selects **A** because $\text{IRR}_{B-A} < \text{MARR}$.